Comparative analysis of Indian Car Retail Market in different BI tools

Module	B9DA106 – Data Visualization
Assessment	Continuous Assessment Two (CA2)
Programme	MSc Data Analytics
Dataset	Car Retail Sales Dataset (23 columns, ~80,000 rows)
Tools Used	Tableau Desktop Public Edition & Power BI
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1. Introduction

This report critically evaluates the design, development, and analytical effectiveness of the data visualization work completed for B9DA106 CA2. The project involved building an interactive sales dashboard in both Tableau and Power BI, creating a novel Sankey diagram and decomposition tree visualization, and comparing the two tools throughout the process. The dataset chosen was a Car Retail Sales dataset comprising approximately 80,000 rows and 23 columns, covering dimensions including date fields (year, month, week, day), dealer identifiers, geographic data (city, region), vehicle attributes (brand, model, category, fuel type, transmission), pricing information (base price, discount, final price), promotional type, inventory, online leads, test drives booked, units sold, and total revenue.

The intended audience for the dashboard is a **Sales Manager** within an automotive retail organization, whose primary need is to monitor sales performance across regions, brands, vehicle categories, and time periods, and to make data-driven decisions around promotions, inventory allocation, and sales team targets. The dataset is rich in both categorical and quantitative dimensions, making it well-suited to this use case.

2. Tool Comparison: Tableau vs Power BI

Both Tableau and Power BI were used to construct near-identical interactive dashboards. Working extensively with both platforms over the course of this project provided substantial insight into the practical differences between the two tools, spanning ease of use, calculated field creation, interactivity, design control, and publishing options.

2.1 Ease of Use and Interface

Tableau's interface centres on a drag-and-drop canvas model that encourages visual exploration. Fields can be dropped onto rows, columns, or mark cards, and the resulting visualization updates in real time. This feedback loop is highly intuitive for rapid prototyping and exploratory analysis. The Marks card system — which controls colour, size, label, detail, and tooltip for each layer — is conceptually elegant once understood.

Power BI adopts a more structured, ribbon-based interface familiar to users of Microsoft Office products. The Fields pane, Visualizations pane, and Filters pane are logically separated, which aids organized development but can feel compartmentalized when compared to Tableau's fluid canvas. Overall, both tools were found to be similarly intuitive once the learning curve was navigated, with Tableau feeling more exploratory and Power BI feeling more methodical.

2.2 Calculated Fields and Data Modelling

Both tools support custom calculations, but their implementation philosophies differ significantly. In Tableau, calculated fields use a concise, SQL-like syntax. Creating measures such as a Total Revenue, Total Units required only a few lines of straightforward arithmetic, and these fields appeared immediately in the data pane for use across any sheet. The ability to create calculated fields on the fly without leaving the visualization context was a considerable productivity advantage.

Power BI uses DAX (Data Analysis Expressions) for calculated columns and measures. DAX is a powerful and expressive language, but it introduces the concept of evaluation context — the distinction between row context and filter context — which adds conceptual overhead. Writing measures such as average revenue per transaction or conditional aggregations required a more deliberate approach. DAX's CALCULATE() function, while powerful, demands an understanding of how filters propagate through the data model. For users with an SQL or Excel background, DAX has a familiar feel, but it is undeniably more verbose than Tableau's formula syntax.

In this project, calculated fields were created in both tools to derive **Total Revenue, Total Units, Total Test Drives, Top Model, Hybrid Fuel Type Share %, PY Revenue**. These measures enriched the dashboard with derived analytical insight not directly available in the raw dataset.

2.3 Interactivity: Filters and Parameters

Tableau's filter and parameter system integrates tightly with the visualization layer. Filters can be applied at the data source, context, or worksheet level, giving fine-grained control over what data each view displays. Parameters in Tableau are particularly powerful — they allow users to dynamically change values (such as

selecting a Top N metric threshold) that drive calculated fields or reference lines across multiple sheets simultaneously.

In Power BI, slicers serve a similar function to Tableau filters and are easy to configure. Cross-filtering between visuals — where clicking a bar in one chart automatically filters all other visuals — is a standout feature that enhances interactivity without any additional configuration. Drill-through pages in Power BI also allow contextual deep-dives into specific data points. In this dashboard, parameters were implemented with a dropdown to select metric as Category, Region, Transmission in both tools, with broadly equivalent results in terms of functional interactivity.

2.4 Design and Aesthetic Control

Tableau provides highly granular control over visual design. Colour palettes can be defined at the worksheet or workbook level, tooltip content is fully customisable, and font, size, and alignment settings are accessible for every text element. The ability to use floating containers in dashboards enabled a precise, layered layout.

Power BI's theming system allows JSON-based theme files to set broad aesthetic rules across the entire report, which is efficient for consistency. However, individual chart-level customisation particularly for axis labels, legend positioning, and data label formatting is less flexible than Tableau. Some formatting options are buried across multiple panes. Despite this, Power BI's conditional formatting feature, which allows colour rules to be driven by data values (e.g., highlighting negative revenue trends in red), was found to be more straightforward than the equivalent approach in Tableau.

Dimension	Tableau	Power BI
Ease of Use	Drag-and-drop is highly intuitive; visual feedback is immediate and exploratory.	More structured ribbon-based UI; steeper initial learning curve but logical once familiar.
Calculated Fields	Formula syntax is concise and SQL-like. Easy to create on the fly.	DAX is powerful but verbose. Measures require understanding of evaluation context.
Interactivity	Filters and parameters integrate seamlessly. Actions between sheets are fluid.	Slicers and drill-through work well; cross-filtering between visuals is a standout feature.
Map Visuals	Excellent built-in geographic mapping with region-level granularity.	Good mapping via Bing Maps; requires precise column categorisation for best results.
Sankey Visuals	Custom Sankey requires a workaround (dual-axis path shapes); no native chart.	Marketplace offers Sankey add-in but limited styling control.
Colour & Design	More granular control over palettes, tooltips, and layout.	Theme JSON offers broad styling; individual chart customisation is more limited.
Cards	In Tableau, we had to build individual sheets for each metric as there is no in-built card feature, which was more time-consuming	Power BI’s multi-card visual allowed for a 'one-click' setup with auto-formatting as M,Bn, K.
Performance	Handles large extracts well; live connection can slow on big datasets.	DirectQuery mode is flexible; Import mode is faster for dashboards.
Page Navigator	Manually engineered Navigation Buttons which allowed for 'Guided Storytelling,' where we control the user's analytical path.	Native Page Navigator, which offers high scalability for corporate reporting

Dimension	Tableau	Power BI
Sankey Diagram	In tableau, Its a drag-and-drop successfully which visualizes the complex relationship between the variables	Requires specific sheet setup and often "Data Densification" or specific extension configurations in power bi.
Decomposition Tree	No decomposition Tree; Built via hierarchies, which makes it clumsy when it has too many details.	This is a native AI visual that allows us to drill down from a regional level—like high sales in the East—directly to the performing model; high visibility and low effort

3. Critical Evaluation of Design Decisions

3.1 Dashboard Design: Strengths

The dashboard was designed with a Sales Manager's analytical workflow in mind. The layout follows a top-down information hierarchy: high-level KPI cards at the top surface the most critical metrics at a glance (total revenue, total units sold, average discount percentage, and revenue per unit), while more detailed charts occupy the lower sections for progressive disclosure of insight.

The use of four distinct chart types was a deliberate decision to ensure that different analytical questions could be answered by the most appropriate visual form. A line chart was used to show revenue trends over time. A bar chart was used for brand-level leveraging the parameter control as category, region and transmission. A donut chart was used for revenue and fuel type breakdowns, appropriate for part-to-whole relationships with a small number of categories. A geographic map enabled spatial reasoning about regional performance, allowing the Sales Manager to identify which cities were under- or over-performing.

The Sankey diagram, included as part of the interactive dashboard beyond the four core charts, was particularly effective in answering the analytical question **"How does revenue flow from vehicle characteristics (fuel type and transmission) to vehicle category and region, and which combinations contribute most to overall revenue?"** surfacing the flow of sales volume from brand through vehicle category to fuel type — a multi-dimensional flow that would require multiple separate charts to convey using conventional approaches.

Colour was applied consistently across both dashboards, with a single accent colour (blue) used for primary metrics and neutral tones for secondary dimensions. This avoided the perceptual overload of rainbow palettes and directed attention to the most important values. Titles on every chart were written as analytical insights rather than labels, improving interpretability for a non-technical manager.

3.2 Dashboard Design: Limitations

One limitation of the dashboard design is that the dashboard was designed primarily for desktop viewing. Neither the Tableau nor Power BI versions were optimized for mobile layouts, which may limit usability if a Sales Manager accesses the dashboard on a tablet or smartphone. Future iterations should incorporate responsive layout design.

3.3 Innovative Visualization: Sankey Diagram, Decomposition Tree

The Sankey diagram was selected as the Task 2 innovative visualization to answer the analytical question: **"How does revenue flow from vehicle characteristics (fuel type and transmission) to vehicle category and region, and which combinations contribute most to overall revenue?"** This is a fundamentally relational and multi-dimensional question that conventional bar or pie charts cannot answer in a single view — they would require either multiple separate charts or complex small multiples, each of which fragments the narrative.

The Sankey diagram addresses this by encoding flow as ribbon width proportional to sales volume, allowing the viewer to simultaneously perceive: which brands contribute the most volume overall (left nodes), how that volume distributes across vehicle categories (middle nodes), and how each category aligns with fuel types

(right nodes). This multi-level decomposition in a single visual representation is significantly more efficient than alternatives.

The decomposition tree is another innovative visualization to answer the analytical question: **"How does revenue distribution vary across regions with different products?"** It acts as a Root-Cause Discovery tool. For deep-dives, the Decomposition Tree is utilized as it directs the path towards the under performing or well performing model. In Power BI, this is a native AI visual that allows us to drill down but in Tableau's version,it's built via hierarchies, which makes it clumsy when it has too many details. Hence Power Bi gives high visibility and low effort with this visualization.

3.4 Addressing the Sales Manager's Analytical Needs

The dashboard was evaluated against the analytical needs of a Sales Manager, whose primary questions are: Which brands and models are selling best? Which regions are underperforming? How are sales trending over time?

The line chart directly answers the temporal trend question. The bar chart answers the brand and model performance question. The geographic map answers the regional question. The KPI cards provide at-a-glance answers to the most common daily questions without requiring any interaction. Filters for year, region, and brand enable the Sales Manager to slice data to their specific territory or time period without needing to interpret data irrelevant to their context.

The Decomposition Tree (available in Power BI) added further analytical depth by allowing the Sales Manager to interactively drill into revenue drivers — for example, starting with total revenue and decomposing by region, then by brand, then by fuel type — mirroring the natural investigative process of an experienced sales analyst. This feature was particularly well received as it provides an open-ended exploration path not constrained by pre-designed chart layouts.

Overall, the dashboard successfully translates a complex, high-volume retail dataset into a coherent, actionable analytical tool aligned with the decision-making context of a Sales Manager. Future improvements would include incorporating forecasting visuals, sales target comparison metrics, and dealer-level performance drill-throughs to further enhance strategic utility.

4. Conclusion

This project successfully delivered interactive dashboards in both Tableau and Power BI using a rich car retail sales dataset of 23 columns and approximately 80,000 rows. The dashboards serve a Sales Manager audience with diverse, appropriately chosen chart types, meaningful calculated fields, and interactive filters and parameters that enhance data exploration.

The comparative experience of working with both tools revealed complementary strengths: Tableau excels in exploratory, fluid visualization development and fine-grained design control, while Power BI offers deep integration with Microsoft's ecosystem, powerful cross-filtering interactivity, and a strong data modelling layer through DAX. Neither tool is categorically superior; the optimal choice depends on the organizational context, data infrastructure, and user familiarity.

The Sankey diagram was the most technically demanding and analytically rewarding component of the project. By encoding a three-level sales flow in a single coherent visual, it addressed an analytical question that conventional charts could not answer without fragmentation, demonstrating both originality and technical implementation quality.

The project reinforced several core data visualization principles: that chart type selection should be driven by the analytical question rather than aesthetics; that progressive disclosure (from summary KPIs to detailed views) aligns with how managers actually consume information; that interactivity should reduce cognitive load rather than add complexity; and that design consistency — through controlled use of colour, typography, and layout — is as important as technical execution. These principles informed every design decision made throughout the project.